Xiyu Ke

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New Brunswick, NJ, USA

Research Interests

Robotic manipulation and robot learning, with a particular interest in learning world models to predict action outcomes and support planning across objects, tasks, and environments.

Education

Rutgers University *Sep. 2025 – Expected May 2027*
M.S. in Electrical and Computer Engineering New Brunswick, NJ
GPA: **3.917/4.00**

Dalian Maritime University *Sep. 2021 – Jun. 2025*
B.S. in Intelligent Science and Technology Dalian, China
Weighted Average: 85.2/100 (Top 20%)

Research Experience

CLEAR: Language-Guided Contact-Constrained Grasping *Jul. 2026 – Present*
Algorithmic Robotics and Control Lab, Rutgers University First author; submitted to ICRA 2027

- Developed a language-guided grasping system for user-specified object parts while avoiding forbidden-contact regions; deployed it on an AgileX Nero arm with a wrist-mounted RealSense D405.
- Integrated reachable close-up RGB-D acquisition with automatic positive and negative point prompting for SAM3, refining fine-grained part segmentation into task-aware contact maps with separate target and forbidden regions.
- Implemented multi-start cross-entropy method (CEM) grasp refinement to optimize target-contact quality while penalizing forbidden-region proximity; enforced hard image-space and 3D contact exclusions, followed by contact and motion-feasibility checks.
- Conducted real-robot evaluations on 16 objects with 160 matched trials per method: CLEAR + AnyGrasp achieved **86.25% task success** versus **68.13%** for the strongest evaluated baseline (SAM3 text + AnyGrasp). Success required complete-object lifting, designated-region contact, and no forbidden contact.

OrganizeIt: Robotic Execution for Tabletop Organization *Jun. 2026 – Aug. 2026*
Algorithmic Robotics and Control Lab, Rutgers University Second author; submitted to AAAI 2027

- Developed real-robot grasping and execution modules, deploying AnyGrasp-based pick-and-place on a UR5e with wrist-mounted RGB-D sensing.
- Designed coordinated arm–gripper motion to compensate for finger extension during closure when picking up thin objects lying flush with the table, including smartphones and rulers; implemented object-specific gripper pre-shaping.
- Implemented force-triggered recovery with re-perception and grasp retries for object tipping, unexpected contact during descent, and missed grasps.
- Incorporated top-down grasp constraints and buffer-placement clearance for cluttered scenes; conducted real-robot experiments, collected data, and analyzed outcomes across four tabletop domains.

microMVP: Vision-Guided Multi-Vehicle Research Platform *Oct. 2025 – Mar. 2026*
Algorithmic Robotics and Control Lab, Rutgers University

- Designed and built modular micro-vehicles with Seeed Studio XIAO controllers, motor drivers, custom PCBs, and 3D-printed chassis; developed Wi-Fi client–server control and ChArUco localization for vehicle position and heading estimation.
- Implemented vision-guided multi-vehicle trajectory execution for circular, figure-eight, and user-drawn paths; produced demonstrations integrating the control interface with physical fleet motion.

Research Experience (continued)

Quadrotor Navigation and Obstacle Avoidance *Apr. 2023 – Jul. 2024*

Dalian Maritime University

- Implemented artificial potential field (APF) obstacle avoidance in CoppeliaSim (V-REP) using Lua; explored ant colony and particle swarm optimization for quadrotor path planning.
- Integrated Crazyflie, distance sensing, ROS-based flight control, and NOKOV motion capture to support indoor navigation and obstacle-avoidance experiments, trajectory inspection, and flight-system debugging.

Publications and Manuscripts

Accepted Conference Paper

[1] Qiuyu Wang, Na Zhao, Chaojun Qin, **Xiyu Ke**, Yudong Luo, and Yantao Shen. (2026). "**Time-Optimal Trajectory Planning and Model Predictive Control of Morphing Quadrotors**." *IEEE International Conference on Robotics and Automation (ICRA 2026)*.

Submitted Manuscripts

[2] **Xiyu Ke**, Litao Liu, Pengfei Yi, Yifan Han, Junshan Huang, Sixu Lin, and Jingjin Yu. "**CLEAR: Contact-Aware Localization and Exclusion-Aware Refinement**." Manuscript submitted to the *IEEE International Conference on Robotics and Automation (ICRA 2027)*.

[3] Junshan Huang, **Xiyu Ke**, Jintong Li, Litao Liu, Duo Zhang, and Jingjin Yu. "**OrganizeIt: Compiling Generated Images into Tidy Tabletop Arrangements**." Manuscript submitted to the *AAAI Conference on Artificial Intelligence (AAAI 2027)*.

[4] Litao Liu, Khushwanth Sri Sai Vudhya, Tanishq Annavaram, Pengfei Yi, Junshan Huang, Yifan Han, **Xiyu Ke**, Sixu Lin, and Jingjin Yu. "**Pushing Clutter via Attention-Based Multi-Object Interaction Dynamics**." Manuscript submitted to the *IEEE International Conference on Robotics and Automation (ICRA 2027)*.

[5] Litao Liu, Pengfei Yi, Enze Yuan, Zilin Yuan, Haoran Du, Ishaan Myrie, Yifan Han, Wenbo Yu, **Xiyu Ke**, and Jingjin Yu. "**DesignAgent: Efficient and Adaptive Search Framework for Robot Morphology Design**." Manuscript submitted to the *AAAI Conference on Artificial Intelligence (AAAI 2027)*.

Technical Skills

Programming & Tools: Python, C/C++, MATLAB; Linux.

Learning & Perception: PyTorch, OpenCV, RGB-D perception, point-cloud processing, semantic segmentation.

Robotics & Simulation: ROS, CoppeliaSim (V-REP), motion planning, grasp pose estimation.

Design & Prototyping: SolidWorks, 3D printing, PCB design.